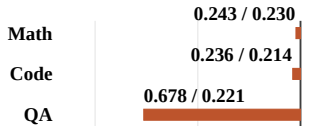
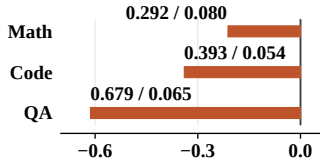


A Pruning

Three checkpoints · power

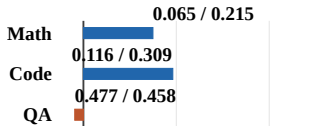


2.8B repeat · power

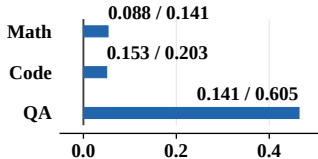


B Grouped quantization

Boundary · interpolation

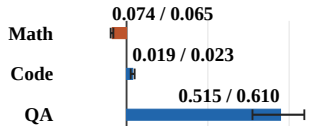


New state · median

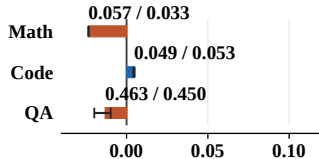


C Distillation

Gemma-3 270M



Gemma-3 1B



MAE(baseline) - MAE(candidate), nats/token; positive = candidate better

Bar-end MAEs: candidate / baseline; whiskers: 95% pool-cluster interval